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Introduction to Data Science — Project Report

Zero-Day Intrusion Detection using Autoencoder

Unsupervised Anomaly Detection

Abstract

This project proposes an unsupervised anomaly detection framework for zero-day intrusion detection using an Autoencoder neural network trained exclusively on benign network traffic. Instead of relying on labeled attack signatures, the system learns normal traffic behavior and flags deviations using reconstruction error. The approach is evaluated on multiple attack categories from the CIC-IDS2017 dataset, demonstrating strong precision for volumetric attacks and room for improvement on subtle, low-intensity intrusions.

1. Introduction

Traditional Intrusion Detection Systems (IDS) rely on known attack signatures or supervised models trained on labeled attack data. Such systems inherently fail when facing zero-day attacks — novel threats that have never been seen before and therefore have no existing signatures.

To overcome this fundamental limitation, this project models normal (benign) network behavior using an Autoencoder neural network. Any traffic that deviates significantly from learned normal patterns is flagged as a potential anomaly. This signature-free approach is capable of detecting previously unknown attack types, making it a powerful complement to traditional IDS systems.

2. Data Collection & Understanding

The CIC-IDS2017 (Canadian Institute for Cybersecurity Intrusion Detection System 2017) dataset was used in Parquet format. This is a widely recognized benchmark dataset for network intrusion detection research.

The dataset spans an entire week of network activity:

- Monday — Contains only BENIGN traffic (used exclusively for model training)
- Tuesday — Bruteforce attacks (FTP-Patator, SSH-Patator)
- Wednesday — DoS/DDoS attacks (Slowloris, SlowHTTPTest, Hulk, GoldenEye)
- Thursday — Web Attacks and Infiltration (SQL Injection, XSS, Brute Force)
- Friday — PortScan, DDoS, and Botnet activity

82 numerical, flow-based features were extracted from raw network captures using CICFlowMeter, including flow duration, packet lengths, inter-arrival times, flag counts, and byte rates. The dataset provides a realistic simulation of enterprise network behavior with labeled ground truth.

3. Data Cleaning & Preparation

Proper data cleaning and preparation were essential for reliable model training. The following steps were applied systematically:

3.1 Label Separation

The Label column was removed from the feature matrix before training. The model is trained in a fully unsupervised manner using only unlabeled feature data from Monday’s benign traffic. Labels were retained separately for evaluation purposes.

3.2 Missing Value & Infinity Handling

Network flow features can sometimes contain NaN or infinity values (e.g., from division-by-zero in rate calculations). These were identified and removed or replaced to prevent numerical errors during training. Only the 82 clean numerical features were retained for modelling.

3.3 Feature Scaling

All 82 features were standardized using StandardScaler (zero mean, unit variance). This step is critical for Autoencoder training as features span vastly different ranges (e.g., byte counts vs. flag counts). The scaler was fit exclusively on training data to prevent data leakage, and then applied identically to validation and all attack datasets during evaluation.

Code used for scaling:

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_val_scaled   = scaler.transform(X_val)
```

3.4 Train / Validation Split

The Monday benign dataset was split into 80% training and 20% validation using a fixed random seed (random_state=42) for reproducibility. This split ensures the model is not evaluated on the same data it was trained on, providing an unbiased estimate of reconstruction error on benign traffic.

4. Exploratory Data Analysis (EDA) & Visualization

Three key visualizations were produced to understand the dataset and validate the anomaly detection setup before model training.

4.1 Class Distribution (Benign Monday)

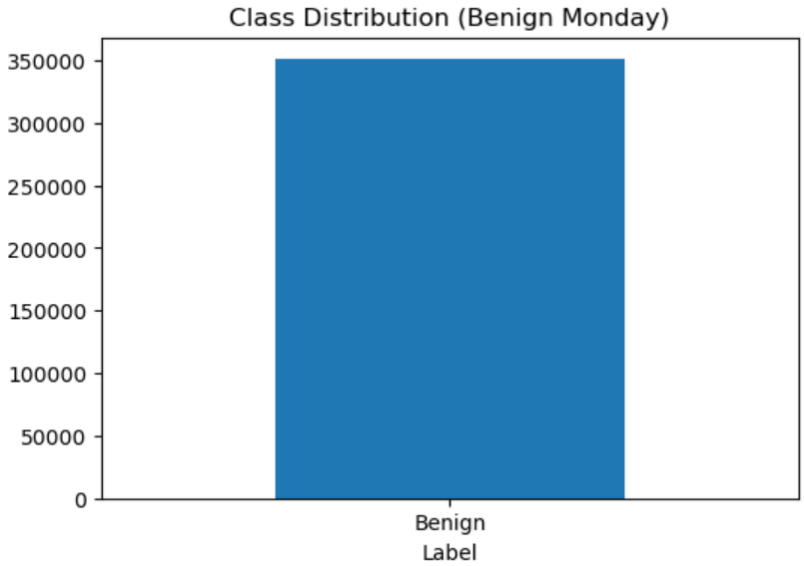

Figure 1: Class Distribution — Monday dataset (Benign only)

Interpretation: The Monday dataset contains exclusively BENIGN samples. This validates the zero-day detection setup: the Autoencoder is trained on purely normal traffic, with no exposure to attack patterns during training. The single-class distribution confirms no label leakage.

4.2 Feature Correlation Heatmap (Benign Data)

Figure 2: Feature Correlation Heatmap on benign traffic (sample of 20,000 flows)

Interpretation: Strong positive correlations are observed among packet length features and inter-arrival time features, confirming that benign network traffic has structured and learnable statistical patterns. These regularities are precisely what the Autoencoder will encode in its bottleneck layer, enabling it to reconstruct normal traffic accurately while failing to reconstruct attack traffic.

4.3 Flow Duration Distribution (Benign vs DoS)

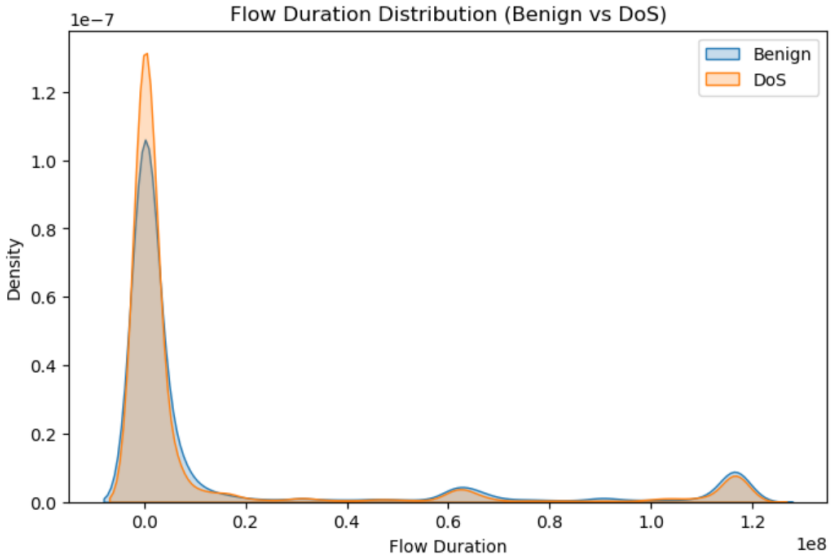

Figure 3: KDE plot comparing Flow Duration distribution for Benign vs DoS traffic

Interpretation: The flow duration distribution of DoS traffic deviates noticeably from benign traffic. DoS flows tend to be shorter and more aggressive, reflecting the flood-based nature of these attacks. This statistical divergence supports the viability of reconstruction-error-based anomaly detection, as the model will struggle to accurately reconstruct these out-of-distribution flow patterns.

5. Model Architecture & Training

An Autoencoder neural network was designed with the following symmetric encoder-decoder architecture:

Input (82) → 64 → 32 → 16 (Bottleneck) → 32 → 64 → Output (82)

All layers use ReLU activation except the output layer which uses linear activation to allow unrestricted reconstruction. The bottleneck layer (16 units) forces the model to learn a compressed, generalizable representation of benign traffic patterns.

Training configuration: loss function = Mean Squared Error (MSE), optimizer = Adam, epochs = 20, batch size = 256. The model is trained exclusively on benign scaled features (X_train_scaled) in a self-supervised manner where input = target output.

5.1 Training vs Validation Loss

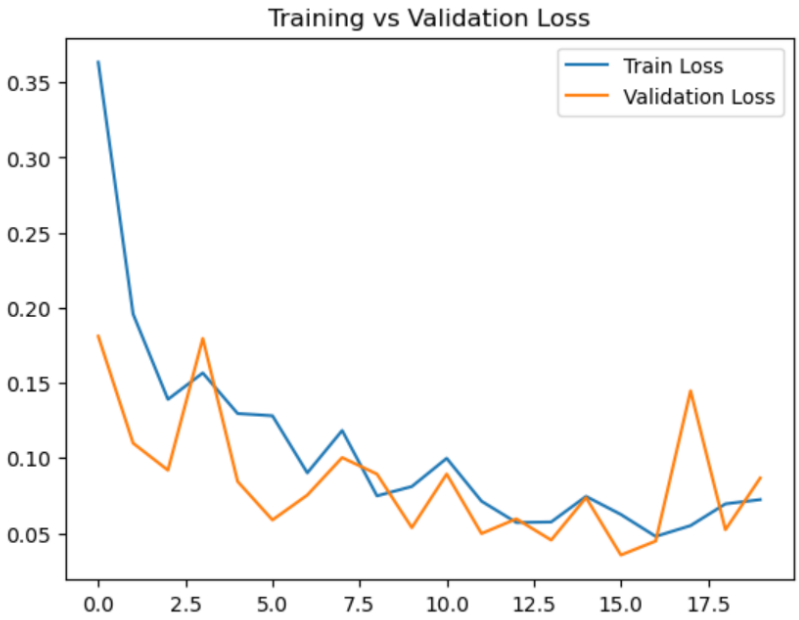

Figure 4: Training vs Validation Loss over 20 epochs

Interpretation: Both training and validation loss decrease steadily and remain close throughout training, indicating stable convergence without significant overfitting. The tight gap between the two curves demonstrates that the model generalises well to unseen benign traffic, a prerequisite for reliable anomaly detection.

6. Anomaly Detection Strategy

The trained Autoencoder reconstructs each input sample and computes the per-sample reconstruction error as Mean Squared Error (MSE) between the input and its reconstruction. Benign traffic, which the model has learned, should have low reconstruction error. Attack traffic, which is out-of-distribution, should exhibit higher reconstruction error.

The anomaly threshold is set at the 90th percentile of the reconstruction error distribution on the benign validation set. Any sample with reconstruction error above this threshold is classified as an anomaly (attack). This percentile-based approach is robust to heavy-tailed error distributions commonly observed in network data.

6.1 Reconstruction Error Distribution (Benign vs Attack)

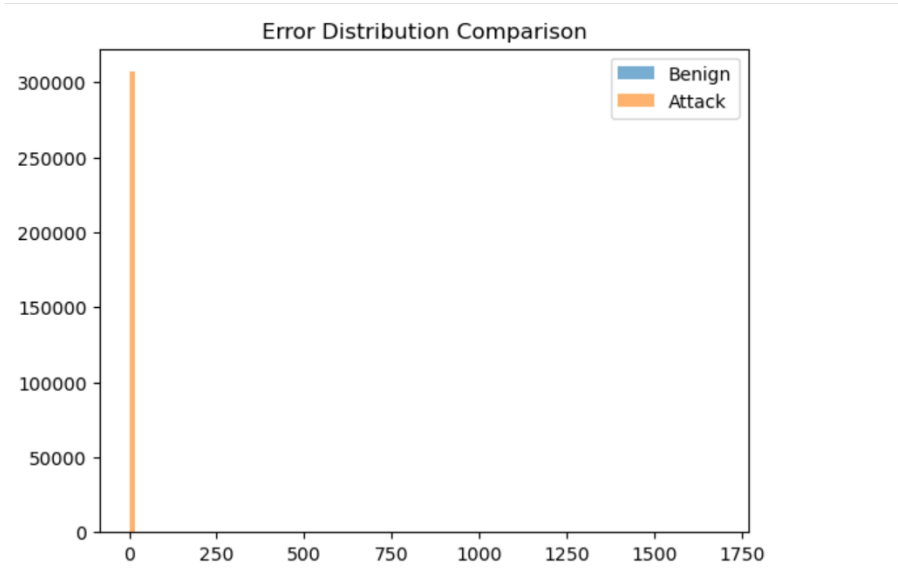

Figure 5: Histogram of reconstruction errors for Benign vs Attack samples

Interpretation: Attack samples generally exhibit higher reconstruction errors than benign samples, validating the core anomaly detection principle. However, subtle attacks (e.g., Bruteforce) partially overlap with the benign error distribution, explaining their lower recall. High-deviation attacks (e.g., DoS, DDoS) are clearly separable.

6.2 Precision-Recall Curve

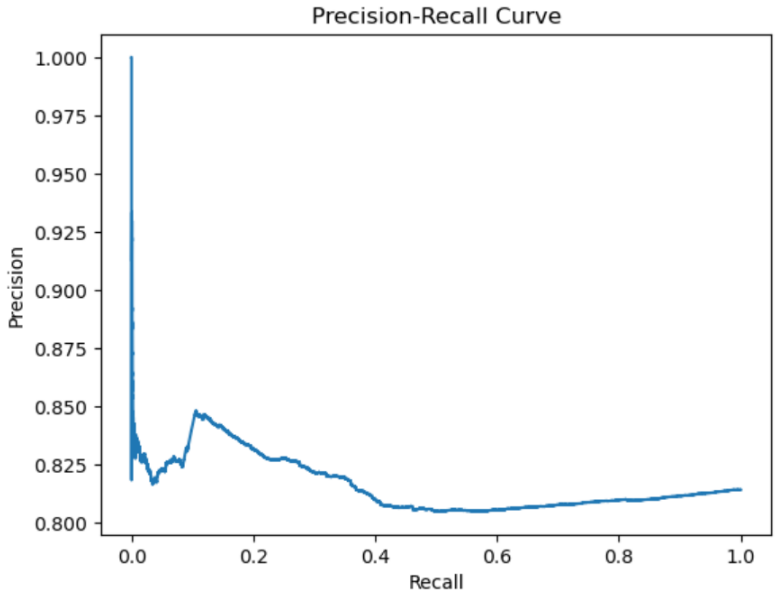

Figure 6: Precision-Recall curve across threshold values

Interpretation: The Precision-Recall curve shows that the model maintains high precision across most threshold values, meaning flagged anomalies are reliably attacks. Recall varies by attack type — volumetric attacks achieve stronger detection while subtle attacks reduce overall recall. The curve demonstrates the precision-recall trade-off inherent in threshold-based detection systems.

7. Results Summary

The model was evaluated against four attack categories from the CIC-IDS2017 dataset using the 90th percentile threshold:

Attack Type	Precision	Recall	Observation
Bruteforce	84%	12%	Low-intensity, subtle
DoS	97%	43%	High-deviation, volumetric
Thursday (Infiltration/Web)	82%	12%	Subtle, evasive attacks
Friday (PortScan/DDoS/Botnet)	95%	33%	Mixed — DDoS well-detected

High Precision = Low False Positives

Low Recall = Misses some subtle attacks

Overall Observation: The Autoencoder excels at detecting high-deviation, volumetric attacks such as DoS and DDoS, where network flow patterns deviate dramatically from normal behavior. Performance is weaker against subtle, low-bandwidth attacks like Bruteforce and Web Attacks, which partially mimic legitimate traffic patterns. The precision scores remain high across all categories, indicating very few false positives — a desirable property for a production IDS system.

8. Limitations

- Low Recall for Subtle Attacks: Bruteforce and web-based attacks that closely resemble normal traffic are difficult to separate using reconstruction error alone.
- Threshold Sensitivity: The 90th percentile threshold is determined from training data and may require tuning in production environments with different traffic profiles.
- Performance Tied to Deviation Magnitude: The model is most effective when attacks cause statistically significant changes in flow features.
- Dataset-Specific Behavior: The scaler and threshold are fit on the CIC-IDS2017 dataset and may not generalize directly to different network environments without retraining.

9. Conclusion & Reproducibility

This project successfully demonstrates an unsupervised, signature-free approach to zero-day intrusion detection using an Autoencoder trained exclusively on benign network traffic. The system achieves strong precision (82–97%) across four attack categories, making it viable for real-world anomaly-based monitoring systems where attack signatures are unavailable.

The key contribution is the ability to detect previously unseen attacks without any labeled attack data during training a direct answer to the limitations of signature-based IDS. The model is particularly effective against volumetric attacks (DoS, DDoS) and can serve as a first-line detection layer, with additional classifiers applied to confirmed anomalies for further categorization.

Future enhancements could include: adaptive thresholding using online learning, hybrid models combining Autoencoders with supervised classifiers for improved recall, and feature engineering to better separate subtle attack patterns.

Reproducibility

All code is implemented in Python using TensorFlow/Keras, scikit-learn, pandas, numpy, matplotlib, and seaborn. The trained model, fitted StandardScaler, and threshold value are saved as artifacts (autoencoder_model.keras, scaler.pkl, threshold.pkl) for reproducible inference on new data. The full pipeline from raw Parquet data to evaluation results is contained in the accompanying Jupyter Notebook (Zero_Day_Intrusion_Detection.ipynb). Random seed 42 is used throughout for reproducibility.

End of Report